EPFL - Computer Science

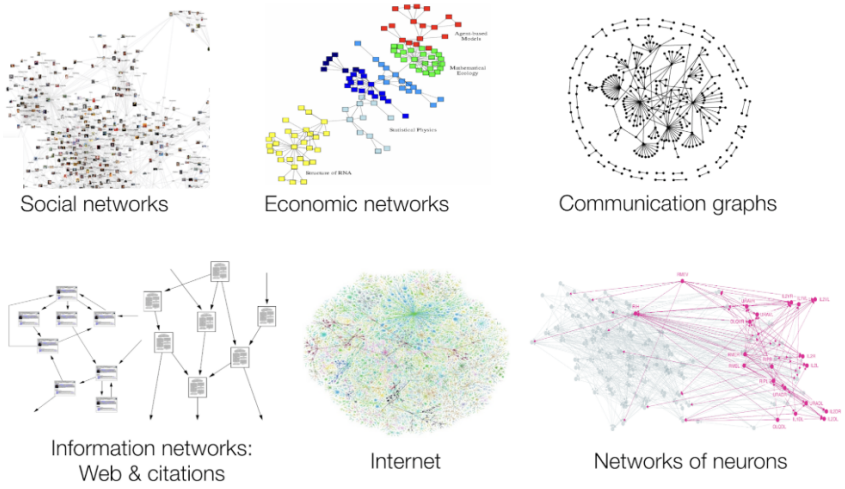
CS-401 Applied Data Analysis

Lecture 13 — 10/12/2025

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Handling network data



Networks as graphs

- **Network:** a real-world system of dependent variables, e.g.,
  - WWW is a network of hyperlinked documents
  - Society is a network of individuals linked by family, friendship, professional ties
  - Metabolic network is a sum of all chemical reactions in a cell
- **Graph:** mathematical abstraction for describing networks
- In practice, ”network” and ”graph” are used interchangeably
- You can make a graph out of almost anything (e.g., connect all people whose name starts with the same letter), so must ask: Does this graph correspond to a meaningful network?

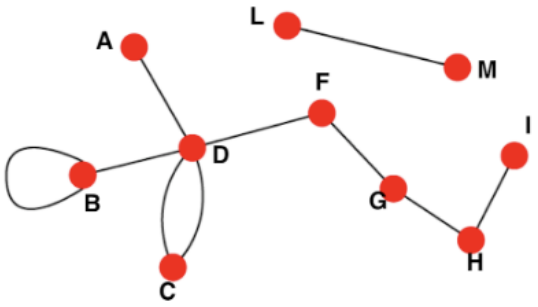
Types of graphs

Annotations

- Entities: nodes/vertices V
- Relationships/interactions: edges/links E
- Entire system: graph $G = (V, E)$
- Node degree, k_i: the number of edges adjacent to node $\boldsymbol{i}$

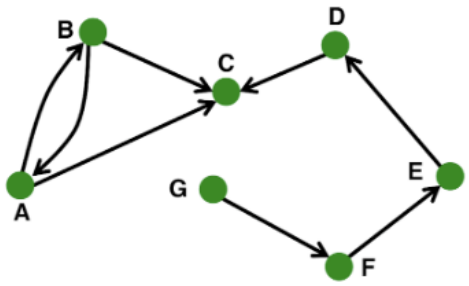
Types of graphs

Undirected
Links: undirected (symmetrical, reciprocal)



Examples:
- Collaborations
- Friendship on Facebook

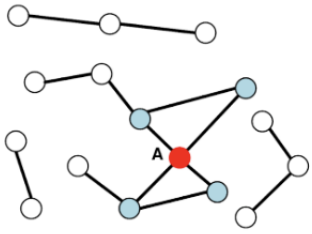
Directed
Links: directed (arcs)



Examples:
- Phone calls
- Following on Twitter

Average node degree

Undirected



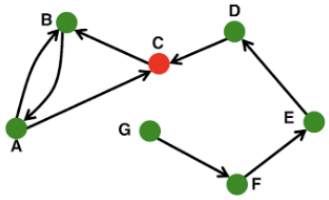
Avg. degree:

$\overline{k} =< k >= \frac{1}{N}\sum_{i=1}^{N} k_i = \frac{2E}{N}$

Intuition: Every edge increases the sum by 2. Every edge increases the degree of 2 nodes.

Directed

In directed networks we define an **in-degree** and **out-degree**. The (total) degree of a node is the sum of in- and out-degrees.

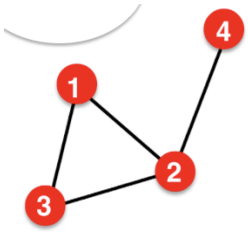


$k_C^{in} = 2 \qquad k_C^{out} = 1 \qquad k_C = 3$

Avg. degree:

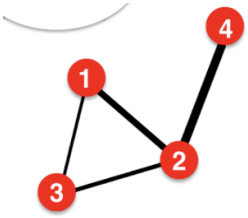
$\overline{k} = \frac{E}{N} \qquad \overline{k^{in}} = \overline{k^{out}}$

Unweighted
(undirected)



Examples: Friendship, Hyperlink

Weighted
(undirected)

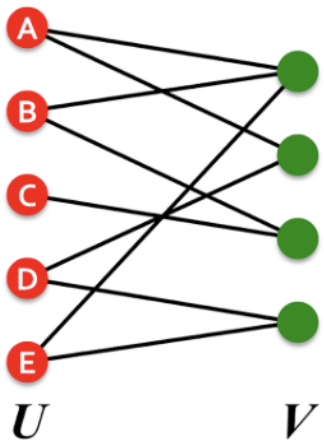


Examples: Collaboration, Internet, Roads

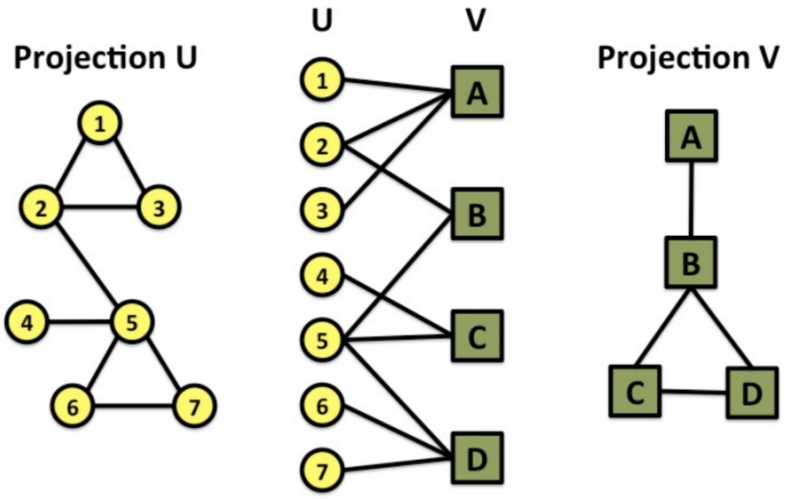
- **Bipartite graph** is a graph whose nodes can be divided into two disjoint sets $\boldsymbol{U}$ and $\boldsymbol{V}$ such that every link connects a node in $\boldsymbol{U}$ to one in $\boldsymbol{V}$

- **Examples:**
  - Authors-to-Papers (they authored)
  - Actors-to-Movies (they appeared in)
  - Users-to-Movies (they rated)
  - Recipes-to-Ingredients (they contain)

- "Folded" networks (a.k.a. "projections"):
  - Author collaboration networks
  - Movie co-rating networks

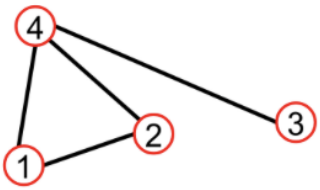


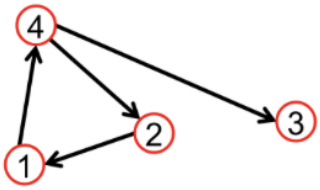
Projects of bipartite graph



Representing graphs on computers

Adjacency matrix





$\mathbf{A_{ij}} = \mathbf{1}$ if there is a link from node $\boldsymbol{i}$ to node $\boldsymbol{j}$

$\mathbf{A_{ij}} = \mathbf{0}$ otherwise

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

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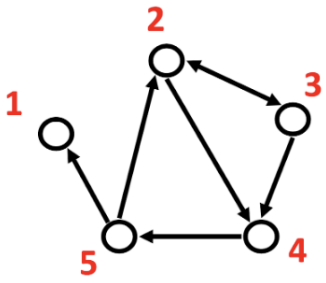
Note that for a directed graph (right) the matrix is not symmetric

But for big graphs... **adjacency matrix is usually sparse**! What is a better representation?

Edge list

Represent a graph as a set of edges

- (2, 3)
- (2, 4)
- (3, 2)
- (3, 4)
- (4, 5)
- (5, 2)
- (5, 1)



Here we only need E entries to store the same information!

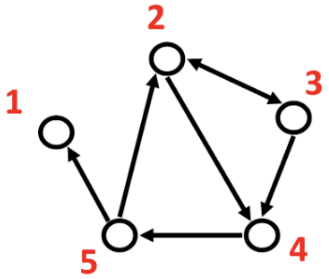
Adjacency list

Adjacency list:

- Easier to work with if network is:
  - Large
  - Sparse

Allows use to quickly retrieve all neighbors of a given node

- 1:
- 2: 3, 4
- 3: 2, 4
- 4: 5
- 5: 1, 2



Properties of real-world networks

- Real networks are different from arbitrary graphs
- Real networks tend to share certain properties
- Remarkable, given the diversity of netwokrs
  - Information networks (e.g., Web graph, knowledge graphs)
  - Social networks (e.g., Facebook, dating networks)
  - Biological networks (e.g., protein-protein interaction)
  - etc...

sparsity

- Every node connected to only small fraction of all nodes
- i.e., $k_i << N$
- Often bounded by a constant
  - e.g., social networks: Dunbar's number (cognitive limit to the number of people with whom one can maintain stable social relationships; allegedly 150)

degree distribution

Degree distribution *P(k)*: Probability that a randomly chosen node has degree $\boldsymbol{k}$

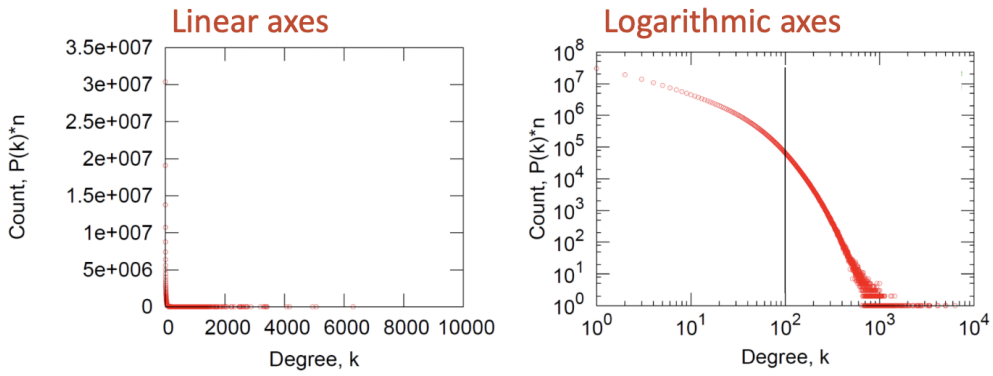
N_k = # nodes with degree $\boldsymbol{k}$

Normalized histogram:

$\boldsymbol{P(k) = \mathbf{N_k} \ / \ N \Rightarrow}$ **plot**

Hence, the degree distribution just means what percentage of nodes has a certain degree $\boldsymbol{k}$.

In a real-world network (social network), it looks more like this:

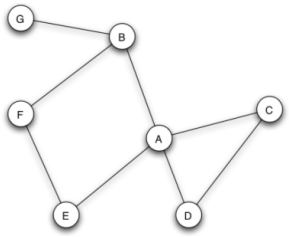


Common behavior in real-world network: power-laws!

If we want to quantify to obtain some information, we should use **median** instead of mean for power laws. Median sometimes is not that informative, so consider using **average of logarithm, exponentiate and thus obtain the geometric mean.**

Triadic closure

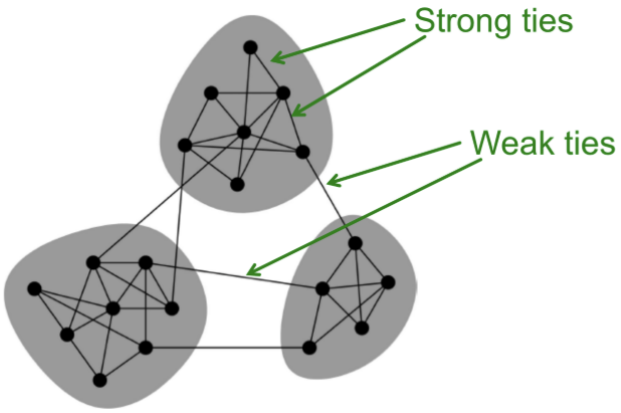
- "A friend of my friend is my friend"
- Measured via clustering coefficient C_i of node i:
  C_i = (# edges among neighbors of i) / (# potential edges among neighbor of i)
- # potential edges amount neighbors of i in undirected graph: $k_i(k_i - 1)/2$



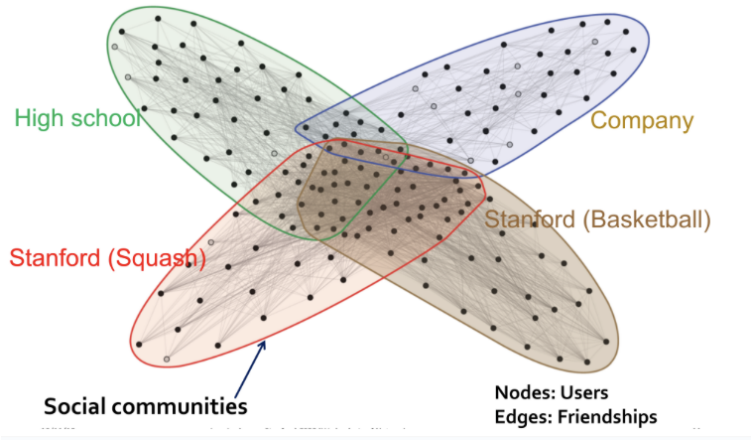
- $C_A = 1/(4 * 3/2) = 1/6$
- In real networks, nodes have **large clustering coefficients**

community structure

- Triadic closure makes real networks **cluster into locally dense "communities"**
- Communities connected via "weak ties"
- Weak ties fill "structural holes"

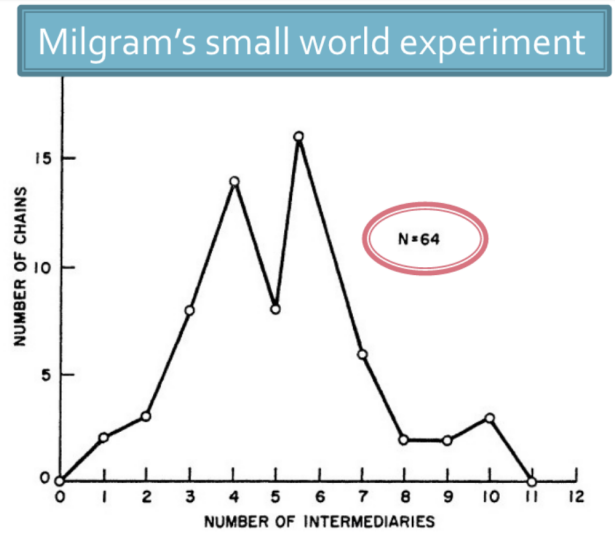


- In real life, communities are often not disjoint, but overlapping:



Average shortest-path length

- **What is the typical shortest path length between any two people?**
  - Experiment on the global friendship network
    - Can't measure, need to probe explicitly
- **Small-world experiment**
  - Picked 300 people in Omaha, Nebraska
  - Ask them to get a letter to a stock-broker in Boston by passing it through friends
- **How many steps did it take?**



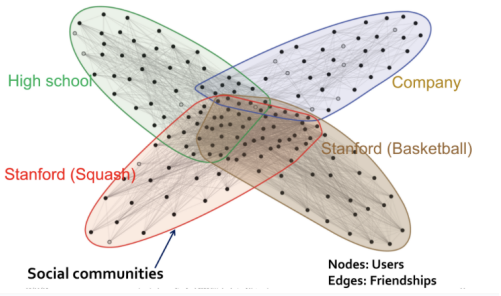
- **64 chains completed:**
  (i.e., 64 letters reaches the target)
  - It took 6.2 steps on the average, thus **"6 degrees of separation"**.
- **Further observations:**
  - People who owned stock had shorter paths to the stockbroker than random people: 5.4 vs 6.7
  - People from the Boston area have been closer paths: 4.4

Navigability

- For decades, people focused on the fact that short paths exist in social networks
- But this is true even in random graphs
- The truly amazing fact is not that short paths exist, but that they are discoverable via greedy decentralized routing (as in Milgram's experiment)
- Example: Wikispeedia: Play a game and see for yourself how well you can navigate an a-priori unknown network.

Homophily/heterophily

- "Birds of a feather flock together"
- Especially in social networks
- Big confound and cause of debate: Influence vs. homophily
- E.g., obese people's friends are more likely to also be obese
  - Influence: I copy eating behavior of those around me
  - Homophily: people with similar eating behavior prone to become friends



Property of real-world networks: summary

Real-world networks (across many types)

- are sparsely connected,
- but some nodes are much more connected than most others (i.e., skewed degree distribution);
- form locally dense clusters via triadic closure,
- which leads to community structure;
- have short paths between random node pairs (partly due to "hubs" [skewed degree distributing!]),
- and the short paths are easily discoverable.

Measure node importance

How to measure "importance" of a node?

- Formalized via *centrality measures*
- Map each node i to a scalar value *C(i)* capturing its importance in the overall network

Degree centrality

- Simplest centrality measure
- Many neighbors ⇒ important node
  - *C(i)* = number of neighbors of i
- Very brittle, easy to "rig"
  - E.g., Twitter: scam account, followed by 100,000 other scam accouns

Closeness centrality

- Farness(x) = total distance to x from other nodes
- C(x) = 1 / Farness(x)

$$C(x) = \frac{1}{\sum_y d(y,x)}.$$

  - Reciprocal to turn farness into closeness
  - Only defined for connected graphs (otherwise d infinite)
- Under closeness centrality, nodes that are easy to reach from anywhere in the network are considerer important
- Variant: harmonic centrality: switch sum and reciprocal

$$C(x) = \sum_{y \neq x} \frac{1}{d(y,x)}.$$

  - C(x) = total reciprocal distance to x to other nodes
  - Defined even for disconnected graphs
    (define *1/d* of disconnected nodes as 0)

Betweenness centrality

- *C(i)* = average fraction of all shortest paths in the network that pass through node i
- Computation of betweenness centrality of node i:
  - For each pair of vertices *(s, t)*:
    * Find all shortest paths from s to t
    * Compute the fraction of these shortest paths that pass through i
  - Average this fraction over all pairs of vertices*(s, t)*
- Expensive to compute

Katz centrality

- Generalization of degree centrality
- Degree centrality counts only number of direct neighbors (i.e., neighbors at distance 1)
- Katz centrality also counts neighbors at distances 2, 3, ...
- More precisely, number of paths from other nodes to i, giving less weight to larger distances:

$$C(i) = \sum_{k=1}^{\infty} \sum_{j=1}^{N} \alpha^k (A^k)_{ji}.$$

> k-th power of adjacency matrix A contains number of length-k paths for each node pair

- More robust than degree centrality

PageRank centrality

- Recursive definition: my centrality $C(i) := x_i$ is high if I receive inlinks from many other central nodes:

$$x_i = \sum_j a_{ji} \frac{x_j}{L(j)}$$

$$L(j) = \sum_i a_{ji}$$

> a_{ji} : entry *(j, i)* of adjacency matrix A (1 if j links to i, else 0)
> *L(j)*: out-degree of j

- Some extra tweaks to make it work with any graph (e.g., disconnected)
- Matrix notation: $\mathbf{x} = \mathbf{M}\mathbf{x}$
  (where M is computed from adjacency matrix A)

- Do you recognize this?
  - x is eigenvector of M with eigenvalue 1 $\Rightarrow$ we're in linear-algebra land
  - x is the steady state the Markov chain induced by the network: x_i is fraction of time a random walker will have spent in node i, after a very long ($\Rightarrow \infty$ random walk
- The technology that made Google huge
  - "Page" in PageRank for Larry PAge
  - MapReduce was invented to compute PageRank on full Google Web crawl
- Bottom line: Pay attention in your linear algebra class